ENG2040 — Data Engineering and Science

Assignment 1: Data Acquisition to Model-Ready Dataset

Marketplace Orders and Delivery Outcomes

September 2026

Format: Individual. **Course weight:** 25%.

Your grade comes entirely from the oral examination, using the weighted criteria below.

Business Goal

The operations team wants to identify orders at risk of late delivery as soon as they are approved. Deliver cleaned order data, business findings and a reusable pipeline that prepares inputs and labels for a future prediction model.

`delivered_after_estimate` is 1 for a usable completed delivery after the estimated calendar date, and 0 otherwise. Delivery on the estimated date is on time; cancelled, unknown or unreliable outcomes remain unlabelled. Use the recorded local clock; no UTC offset is supplied.

Assume basket contents, prices and freight, estimated delivery date, customer location and product/seller attributes are available at approval. Payments, final status, shipment/delivery events, reviews and `shipping_limit_date` cannot be predictors.

Database setup and brief data notes are in Appendices A and B.

1 Loading the Data (20%)

Access the supplied data efficiently and assemble one row per order. What do the tables and keys represent? How do you preserve records, counts and totals? Why is your approach the most efficient suitable choice in runtime, memory and storage?

2 Data Quality and Cleaning (20%)

Profile and clean the data to produce reliable order records and delivery labels, with traceable changes and exclusions. Which problems did you find and address, and why? How do unresolved issues and exclusions affect your results?

A missing optional catalogue field alone does not make an order unusable.

3 Exploratory Analysis (30%)

Use summaries and charts to answer the following business questions about delivery delays.

1. How often do orders arrive late, and how long are the delays?
2. How does the time from approval to the estimated delivery date relate to late delivery?
3. How do late-delivery rates vary across customer states, and where should the operations team investigate first?

How does data quality affect your answers? How sensitive are they to which records you include, and how much confidence should the operations team place in your findings?

4 Feature Engineering (20%)

Create a numerical feature table to support a prediction model for late deliveries. Use one row per retained order and only information available at approval. How did you represent numerical, categorical and date/time fields? What new features did you create? Which features did you keep, and why?

Explain what each feature means. Include matching order IDs and reliable delivery labels. Keep order IDs and labels separate from the features.

5 Reproducibility and Documentation (10%)

Can another person reproduce the **output data from every pipeline step** with one command? How do estimated and measured resource needs compare at each step, including peak memory, temporary and saved storage, intermediate data, and runtime? Does your **6–8 page report (excluding references)** justify your decisions, findings and limitations?

Include chart data, code and any executed SQL, environment details, checks, and a README mapping each step to its inputs, outputs, row meaning and loading instructions; choose formats freely. Findings concern this modified extract; associations do not establish causation.

Grading Rubric

The oral examination uses the following criteria and weights. The questions in this sheet form the basis of the oral examination, but the examination is not limited to them.

Efficiency is assessed throughout. Excellent work justifies its choices against reasonable alternatives and explains estimated and measured memory and storage requirements at every pipeline step, with units and measurement scope. Assessment considers the workload, environment and correctness; no particular tool is required.

Loading the Data (20%)

Excellent	**Adequate**	**Insufficient**
Efficiently accesses and assembles the data, with evidence that record meanings, relationships and totals are preserved.	The data are usable, but relationship checks or efficiency comparisons need clearer evidence.	Missing records, incorrect relationships or unexplained resource use undermine the work.

Data Quality and Cleaning (20%)

Excellent	**Adequate**	**Insufficient**
Supports cleaning decisions with evidence, preserves uncertainty and valid exceptions, and accounts for every record.	Addresses the main issues, but some decisions or audit records need more support.	Unsupported changes, unexplained record loss or unreliable labels undermine the work.

Exploratory Analysis (30%)

Excellent	**Adequate**	**Insufficient**
Answers the questions with suitable evidence and explains limitations and uncertainty or sensitivity.	Findings are useful, but their justification or limitations need more explanation.	Analyses are missing, summaries are misleading or claims lack evidence.

Feature Engineering (20%)

Excellent	Adequate	Insufficient
Features suit the task and available information. Evidence supports their reliability and freedom from leakage.	Features are appropriate, but some choices or checks need more explanation.	Features leak information, use unavailable inputs or produce inconsistent representations.

Reproducibility and Documentation (10%)

Excellent	Adequate	Insufficient
A fresh run reproduces the data at every step, and the documentation makes decisions easy to trace.	Reproducing the work requires minor fixes or clearer instructions.	The outputs cannot be reproduced or traced back to the input.

Dataset Source

This is a modified version of Olist's Brazilian E-Commerce Public Dataset, for educational purposes only. Source licence: CC BY-NC-SA 4.0.

Appendix A: Database setup

The dataset is supplied as `data/raw.sql`. The script `setup_db.py` creates and populates a local SQLite database from it. Python 3.10 or later is sufficient; no additional packages or database server are needed.

From the extracted assignment folder, run:

```
python3 setup_db.py
```

The result is `marketplace.sqlite`, which you can open with a SQLite client or access from your own code. The script leaves an existing database unchanged; use `--db another_name.sqlite` to create another working copy.

The loader imports the supplied records without cleaning them. The tables have SQL column types and primary keys, but their contents still require investigation. Appendix B provides a few essential data meanings.

Appendix B: Data notes

- **Amounts:** `price` and `freight_value` are the price and freight attributed to each item, in Brazilian reais (BRL). `payment_value` is in centavos: 100 centavos = 1 BRL.
- **Customers:** `customer_id` identifies a customer record for an order. `customer_unique_id` links the same customer across orders.
- **Items:** Each item row represents one purchased item. `order_item_id` is its sequence within the order.
- **Payments:** Each `payment_value` is the amount for that payment record, not per installment. Zero installments can be valid.